Self-Supervised Representation Learning for Small-Data Scene Recognition

ITCS 6169/8169 Assignment 2 · STUDENT NAME · https://github.com/USERNAME/REPOSITORY

Best test accuracy: 90.75% (363/400). Ensemble of six ResNet-18 networks (three DINO-pretrained, three SimCLR-pretrained), each pretrained and fine-tuned from random initialization on the 2,400 provided training images only; no external data or weights are used. The repository contains all code, the six FP16 checkpoints, and the commands for every experiment.

1 Setup and Evaluation Protocol

The data consist of 2,400 training images (150 per class, 16 classes) and 400 test images. Fifteen classes are grayscale; *Flower* is the only RGB class. All model selection used a fixed, class-stratified split of the training images into 1,920 training and 480 validation images (exact duplicates kept on one side). For every comparison, self-supervised pretraining used only the 1,920 training-split images; validation and test images never enter a loss. Images are decoded once to 256×256 and augmented on the GPU; networks receive 224×224 inputs (128 and 256 px are studied as controls). Unless stated otherwise the network is a torchvision ResNet-18 with `weights=None`. Training budgets are fixed and the final-epoch model is reported (no early stopping), so the final configuration can be retrained on all 2,400 images without a validation set. Representation quality is measured on frozen features with a weighted cosine kNN classifier (k = 20) and a linear logistic-regression probe. The test set was evaluated only after the final configuration had been fixed on validation.

2 Supervised Baseline

A randomly initialized ResNet-18 is trained with cross-entropy only for 200 epochs: SGD with Nesterov momentum 0.9, learning rate 0.1 (5 warm-up epochs, cosine decay), batch size 128, weight decay 5×10^{-4}, label smoothing 0.1, and random resized crops (35–100% of the area), horizontal flips and mild brightness/contrast/saturation jitter. It reaches 85.0 ± 0.4% validation and 83.7 ± 0.4% test accuracy. Exactly the same recipe is used to fine-tune every pretrained encoder, so differences between rows of Table 1 isolate the effect of the initialization.

3 Self-Supervised Learning Strategy

The main strategy is two-stage: 800 epochs of self-supervised pretraining without labels, after which the projection head is discarded and all layers are fine-tuned with the baseline recipe; joint optimization is evaluated as an alternative.

DINO (self-distillation). A student and a teacher network of identical architecture map an image view to a probability distribution over K learned prototypes (three-layer MLP head with 2,048 hidden units, a 256-d L2-normalized bottleneck, and unit-norm prototype vectors). The teacher receives no gradients; its weights are an exponential moving average of the student's (momentum 0.996 increasing to 1). For each image, two global crops (224 px, 25–100% of the area) and six local crops (96 px, 5–25%) are sampled. The teacher sees only the global crops, and the student is trained to predict the teacher's output for one global crop from every other crop by minimizing the cross-entropy between the teacher distribution (temperature 0.04, after subtracting a running mean of its outputs, the *center*) and the student distribution (temperature 0.1). Centering prevents a single prototype from dominating and the low teacher temperature prevents the uniform solution, so collapse is avoided without negative pairs. The local-to-global constraint requires a small region of a scene (e.g., part of a bed) to predict the representation of the whole image, which suits categories defined by their constituent objects and layout. Adaptations to this dataset: K = 1,024 prototypes instead of 65,536, since there are only 1,920 images; a fixed teacher temperature without warm-up; batch size 64 (30 updates per epoch); SGD with learning rate 0.05 and weight decay 10^{-4}. The teacher backbone is used downstream.

SimCLR (contrastive learning). Two strongly augmented views of each image (random resized crops covering 8–100% of the area, color jitter, random grayscale, Gaussian blur) are embedded through a two-layer projection head (128-d); the NT-Xent loss (temperature 0.2) must identify the matching view among the 511 other views in a batch of 256. **Joint training** instead adds 0.5 × the SimCLR loss, computed on the same mini-batch and backbone, to the cross-entropy loss during the 200-epoch supervised schedule from random initialization.

4 Results and Analysis

Method	kNN	Linear	Val. acc.	Test acc.	Gap
ResNet-18, 224 px					
Supervised baseline (random init.)	–	–	85.0 ± 0.4	83.7 ± 0.4	15.0
Supervised + MixUp/CutMix	–	–	87.9 ± 0.9	83.7 ± 1.4	12.1
Joint CE + 0.5·SimCLR (random init.)	–	–	86.6 ± 0.6	82.6 ± 1.0	13.4
SimCLR → fine-tune	80.6	82.5	88.8 ± 0.8	85.2 ± 0.7	11.2
SimCLR → fine-tune + MixUp/CutMix	–	–	89.8 ± 1.0	87.8 ± 0.1	10.2
DINO → fine-tune	82.9	83.1	**90.8 ± 0.7**	**88.0 ± 0.5**	9.2
DINO → fine-tune + MixUp/CutMix	–	–	89.4 ± 1.3	88.3 ± 0.3	10.6
ResNet-18, resolution control (256 px is the native size of half of the images)					
128 px: supervised baseline	–	–	81.0 ± 1.2	80.1 ± 0.9	19.0
128 px: SimCLR → fine-tune	80.4	78.8	85.5 ± 1.1	80.9 ± 0.9	14.5
128 px: DINO → fine-tune	81.5	83.5	87.4 ± 2.0	84.3 ± 0.6	12.6
256 px: supervised baseline	–	–	84.6 ± 0.6	84.5 ± 1.1	15.4
256 px: SimCLR → fine-tune	80.6	84.2	89.9 ± 0.3	86.0 ± 0.8	10.1
256 px: DINO → fine-tune	79.6	84.0	90.0 ± 0.8	89.3 ± 0.8	10.0
ResNet-50, 224 px (backbone control)					
Supervised baseline	–	–	82.2 ± 2.0	80.5 ± 2.4	17.6
Supervised + MixUp/CutMix	–	–	86.9 ± 1.6	84.7 ± 4.3	12.9
DINO → fine-tune + MixUp/CutMix	81.0	82.7	90.1 ± 0.1	89.1 ± 1.0	9.9
Final system, 224 px (flip test-time augmentation)					
Ensemble: 3 × DINO → FT + 3 × SimCLR → FT+MixUp, trained on 1,920 images			92.5	88.5	–
Same ensemble, SSL and fine-tuning on all 2,400 images (submitted)			–	**90.75**	–

Table 1. Accuracy (%). Except for the final-system rows: single models, mean ± std over three fine-tuning seeds, no test-time augmentation. kNN/Linear: frozen pretrained encoder, validation split. Gap: training minus validation accuracy. All rows except the last use the 1,920-image training split.

SSL improves accuracy with 2,400 images. With an identical fine-tuning recipe, DINO initialization improves on random initialization by 5.8 points on validation and 4.3 points on test, and SimCLR by 3.8 and 1.6 points. When both are regularized with MixUp/CutMix, the SSL-initialized models are 1.5–1.9 points better on validation and 4.2–4.7 points better on test. On the test set MixUp/CutMix does not improve the randomly initialized model (83.7% in both cases) but improves both SSL-initialized models, which suggests that strong regularization and SSL pretraining are complementary rather than redundant.

SSL reduces overfitting. All models fit the training set almost perfectly, but the train–validation gap falls from 15.0 points (random initialization) to 11.2 (SimCLR) and 9.2 (DINO), and the validation negative log-likelihood falls from 0.60 to 0.49 (DINO). The pretrained encoders already separate the classes to a large extent (83% linear-probe accuracy), so the labels are used to refine rather than to create the features.

Pretraining followed by fine-tuning outperforms joint training. Adding 0.5 × SimCLR to the supervised loss improves validation accuracy by 1.6 points but lowers test accuracy by 1.1 points, well below SimCLR pretraining followed by fine-tuning (+3.8 / +1.6). In joint training the two objectives compete for the same features from the first step, and the SSL objective receives 200 instead of 800 epochs, so the comparison is not compute-matched. The result is nevertheless consistent with the benefit of SSL in this setting coming mainly from a better initialization rather than from SSL acting as a regularizer.

A more complex method helps, but less than its cost suggests. DINO takes about 2.4× the wall-clock time of SimCLR (24 vs. 10 min) and has more hyperparameters. Its frozen features are only slightly better (kNN 82.9 vs. 80.6; linear 83.1 vs. 82.5), and its fine-tuning advantage (+2.0 validation, +2.8 test) shrinks to within seed variance once MixUp/CutMix is added (89.4 vs. 89.8 validation; 88.3 vs. 87.8 test). A plausible reason for SimCLR's weaker plain fine-tuning is class collision: with 16 balanced classes, about 6% of the 510 negatives of each view are images of the same class, which the loss pushes apart. DINO uses no negative pairs.

SSL saturates early. DINO's kNN accuracy reaches 82.5% after 250 of 800 epochs and then remains between 81% and 83%, while the loss keeps decreasing from 1.36 to 0.15 nats. Because the cross-entropy is an upper bound on the entropy of the teacher's output, the teacher's distribution over prototypes becomes nearly one-hot for each image, which suggests that further training mainly memorizes individual images. With 1,920 images, the amount of data rather than the training length limits the learned representation.

Resolution and backbone. Increasing the input size from 128 to 224 px improves every method by a similar amount (3.3–4.0 validation points, 3.6–4.3 test points); the SSL gain is approximately additive to the resolution gain (DINO: +6.5 / +4.2 at 128 px, +5.8 / +4.3 at 224 px). Going further to 256 px, the native size of half of the images, changes validation accuracy inconsistently (−0.8 to +1.7 points over seven configurations, including joint training and MixUp/CutMix; see repository) but raises single-model test accuracy by 0.5–1.5 points for every configuration; since the two small evaluation sets disagree, we cannot conclude that 256 px is better than 224 px. Frozen-feature quality depends little on resolution (DINO linear probe 83.5, 83.1 and 84.0 at 128, 224 and 256 px). Trained from random initialization, ResNet-50 is worse and less stable than ResNet-18 (82.2 ± 2.0 vs. 85.0 ± 0.4 validation). DINO pretraining improves it by a similar margin as ResNet-18 and markedly reduces its sensitivity to the seed (test standard deviation 4.3 → 1.0 with MixUp/CutMix), yielding the highest single-model test accuracy among the 224 px configurations (89.1%). Its frozen features are nevertheless not better than ResNet-18's (kNN 81.0, linear 82.7), again pointing to data as the limiting factor.

Final system. The two objectives make partly different errors: an ensemble of the three DINO models reaches 91.3% validation accuracy, and an ensemble of these models with the three SimCLR + MixUp/CutMix models reaches 92.5%. This ensemble was selected on validation among three ensembles specified in advance (92.5%, 92.1%, 90.6%), before the test set was used. Repeating the selected configuration, including SSL pretraining, on all 2,400 images raises its test accuracy from 88.5% to 90.75%. The 256 px study was run after this test evaluation and subjected to the same rule: its best ensemble reached 91.7% validation accuracy (vs. 92.5%), so the submitted system was not changed and the 256 px ensemble trained on all images was not evaluated on test.

5 Limitations

Statistical resolution. All results use one split with 480 validation and 400 test images (one image corresponds to 0.21 and 0.25 points). Several differences discussed above are within seed variance, and validation accuracy exceeds test accuracy for every configuration (by up to 4.9 points). Each SSL configuration was pretrained once, so the reported standard deviations reflect only fine-tuning variance. **Tuning and controls.** The DINO hyperparameters (1,024 prototypes, teacher temperature 0.04) were taken from a preliminary low-resolution exploration and not re-tuned; SSL augmentations were not ablated; the joint training comparison is not compute-matched. The submitted models have no held-out validation set, so their accuracy can only be inferred from the 1,920-image counterparts. **Dataset artifacts.** *Flower* is the only RGB class, and image size is class-dependent (e.g., all *Coast* images are 256×256 and all *Suburb* images 330×220); after stretching to a square, both color and aspect distortion are potential shortcuts. Only convolutional backbones were studied; masked image modeling with Vision Transformers was not evaluated.

6 Conclusion

Self-supervised pretraining on the same 2,400 images is clearly beneficial in this limited-data regime. Under an identical fine-tuning recipe it improves ResNet-18 test accuracy by 1.6–4.3 points, reduces overfitting, and stabilizes a larger backbone. The benefit is obtained by pretraining followed by fine-tuning rather than by joint optimization, is approximately independent of input resolution, and is largest for self-distillation without negative pairs. At the same time, frozen-feature quality saturates after a few hundred epochs and improves little with a larger backbone or a higher resolution, so the amount of data, rather than the choice of objective, is the main limitation. AI coding assistants (Claude Code) were used to implement the pipeline, run experiments and draft this report; see AI_USAGE.md in the repository.